SYSTEM ADMINISTRATOR LAB

DAY 05: NTDS.dit & SYSVOL

Windows Server 2025 • Domain: lab.local • DC: LAB-DC01

Objective: Learn where AD stores directory data, where domain policy files and scripts live, and how to verify both safely.

1. Understand the difference

NTDS.dit	The main Active Directory database file. Stores directory objects and attributes such as users, groups, computers and other AD information.
SYSVOL	A shared folder containing domain Group Policy files and logon/startup scripts.

2. Verify NTDS.dit

Run PowerShell as Administrator on LAB-DC01. The registry query shows the configured database path:

```
Get-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Services\NTDS\Parameters" | Select-Object "DSA Database file"
```

Check whether the configured file exists:

```
$dbPath = (Get-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Services\NTDS\Parameters")."DSA Database file"
Test-Path $dbPath
```

Expected: True. This confirms that the path exists, not that the database is fully healthy. Default path is commonly C:\Windows\NTDS\NTDS.dit, but it can differ.

3. Verify SYSVOL and NETLOGON

List published shares:

```
net share
```

Look for SYSVOL and NETLOGON. Then check the local domain folder:

```
Test-Path "C:\Windows\SYSVOL\sysvol\lab.local"
Get-ChildItem "C:\Windows\SYSVOL\sysvol\lab.local"
```

Test access through the network shares:

```
dir \\LAB-DC01\SYSVOL
dir \\LAB-DC01\NETLOGON
```

4. Inspect Group Policy files

```
Get-ChildItem "C:\Windows\SYSVOL\sysvol\lab.local\Policies"

Get-ChildItem "C:\Windows\SYSVOL\sysvol\lab.local\Policies" -Filter "gpt.ini" -Recurse
```

A GPO has two related parts: Group Policy Container (GPC), stored in Active Directory, and Group Policy Template (GPT), the file-based portion stored in SYSVOL. A gpt.ini file contains policy-template version information. Do not edit these files manually.

5. SYSVOL replication

Modern Windows Server domains normally use DFS Replication (DFSR) to replicate SYSVOL between domain controllers. Your lab currently has one DC, so there is no second DC to test replication against.

6. Troubleshooting checklist

Symptom	Check first
NTDS.dit path unreadable	Use elevated PowerShell; confirm AD DS is installed and inspect the configured path.
SYSVOL share missing	Check AD DS/SYSVOL readiness and event logs; do not recreate folders manually.
NETLOGON share missing	Check DC initialization and advertising health.
Network share fails	Check name resolution, connectivity, share availability and permissions.
GPO not applying	Check client domain membership, DNS, GPO scope, permissions and Group Policy logs.
Internet unavailable	Your isolated lab has no gateway by design; this alone does not indicate a SYSVOL problem.

7. Practice task

- Run the NTDS.dit registry query and record the path.
- Run Test-Path against that path.
- Use net share and confirm SYSVOL and NETLOGON.
- Check the local lab.local SYSVOL folder and Policies folder.
- Search for gpt.ini files.
- Run dir \\LAB-DC01\SYSVOL and dir \\LAB-DC01\NETLOGON.
- Record any errors exactly; do not change permissions or delete files as a first response.

8. Interview quick revision

What is NTDS.dit? The main AD DS database file.
What is SYSVOL? A shared location for Group Policy files and scripts.
GPC vs GPT? GPC is in Active Directory; GPT is the file-based policy portion in SYSVOL.
Does seeing SYSVOL in net share prove everything is healthy? No; it confirms the share is published, but further checks may be needed.
Can you manually edit NTDS.dit? No; use supported AD management and backup/recovery tools.

Safety note: These are inspection-only exercises. Do not modify NTDS.dit, recreate SYSVOL manually, or delete policy folders.